

Exam examples

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1 Problem 1

1.1 Description of the problem

We are dealing with a CT scanner¹.

We have the following energies:

$$\varepsilon_{eff} = \varepsilon_{before} \alpha^{\Delta_i} \quad (1.1)$$

where α represent the opacity (so the density of the body, a constant for the problem).

The total energy is given by

$$\varepsilon_0 \prod \alpha_i^{\Delta_i} = \varepsilon_n \quad (1.2)$$

it's exponentially modulated by Δ_i . We can easily get rid of the product by considering the logarithm:

$$\log(\varepsilon_0) + \sum \log(\alpha_i) + \Delta_i \quad (1.3)$$

Assume we know the density of each part of the body, can we determine the energy density of the image?

Assumed that the position of the emitter is known, Δ_i is constant. But what it's Δ_i ? The friction of a pixel that it's intersected by a fan when the emitter is located at a certain (known) angle and considering the pixel. If a pixel is occluded, $\Delta_i = 0$.

For every angle of the emitter (of machinery) θ_k and for every pixel on the grid $x_{i,j}$, I can calculate the amount of that pixel that will be visible in the beam $\Delta_{i,j,k,q}$:

- i, j index of the pixel on the grid,
- k position of the emitter,
- q indicates the index of sensing element on linear camera (that the pixel can be visible in different regions).

This can be precalculate since the geometry of the environment is known and bound to repeat itself.

1.2 Solution

When we take a CT scan, we take a series of measurement that will impress the pixel q :

$$z_{q,k} = \log(\varepsilon_0) + \sum_{i,j} \log \alpha_{i,j} \Delta_{i,j,q,k} \quad (1.4)$$

We now solve it for least square. What is the state? So what is the thing that can help us predict the results if we know it?

The state is the density of every pixel $x_{i,j} = \log(\alpha_{i,j})$ and has dimension R^n , where n is the number of pixels.

The measurement is $z_{q,k}$. We can re-write the measurement to be

$$z_{q,k} = E_0 + \sum \Delta_{i,j,q,k} x_{i,j} \quad (1.5)$$

and have dimension equal to the sensing element.

How many least square iterations we need? 1 since the error is linear in x , so the solution comes down to a matrix multiplication.

¹"A TAC" - G.Grisetti, 09/12/2024

²For all pixels

2 Problem 2

Assume we have a unicycle/differential drive robot. We know the position of the landmark (which is uniquely identified). The sensor (of unknown position) is measuring the bearing angle from the sensor to the landmark.

2.1 Solution

First step: determine state, controls and measurement.

The state (things we don't know but need to know to determine the measurement) is formed by:

- the position and orientation of the robot (defined in $SE(2)$),
- position of the sensor (defined in $SE(2)$)
- the kinematic parameter: the dimension of the wheels k_l, k_r and the baseline b . Defined in R^3

The measurement is the bearing angle z^k , so it has dimension $SO(2)$. The controls are:

- encoder ticks of the wheel are in the control (so in the prediction) $u \in R^2$

To simplify calculations, knowing that it's actually $SE(2)$, we will consider some simplifications ("Thor's hammer". You can do this in the exam, just make sure you specify that you're doing this to simplify the calculations and that you know the proper/correct dimensions):

- $x^r \in R^3$ for the pose of the robot,
- $x^s \in R^3$ for the pose of the sensor,
- $x^o \in R^3$ for the kinematic model/odometry.

Second step: transition function

Since we're talking of control, we are working with filters.

$$x_t, u_t \rightarrow x_{t+1} \quad (2.1)$$

$$\begin{aligned} x_{t+1}^o &= x_t^o \\ x_{t+1}^s &= x_t^s \\ x_{t+1}^r &= kin(x_t^o, x_t^r, u_t) \end{aligned} \quad (2.2)$$

Where kin indicates that the next position of the robot depends on its kinematics.

Third step: measurement To determine the measurement we simply consider

$$atan2(x_r^{-1}x_s^{-1}p) \quad (2.3)$$

3 Problem 3

3.1 Description of the problem

We are on a world in the middle of the universe. On the surface of the world there is a person with neck problems that can only look up, this poor unfortunate soul is using a camera to watch the star. The stars are far away element of the sphere and their position can be determined considering their position on the sphere S^2 .

3.2 Solution

Step 1: determine the state

The state is formed by:

- the pose of the person $x^r \in SO(3)^n$, (the power of n is because we can move on the planet, so we have a trajectory - least square),
- the position of the stars $x^s \in (S^2)^m$.

Step 2: observation function

Assume we know the pose of the person/robot and the unit vector of the star, what will we measure?

$$h_{n,m}(x_m^s, x_n^r) \quad (3.1)$$

We have our position and or unit vectors that points to the star, the goal is to define the position of the star in our reference frame. How to do that? A simple rotation matrix:

$$R_n^T x_m^s \quad (3.2)$$

we defined a $\boxplus$ operator that defines an orientation on the x axis and on the y axis. This is because we need to keep in mind perturbations Δ_z, Δ_y (rotations along z and y, respectively):

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \boxplus \begin{pmatrix} \Delta_z \\ \Delta_y \end{pmatrix} = R_z(\Delta_z) R_y(\Delta_y) \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (3.3)$$

We defined the function $v2r$ analogue to what we did for $v2t$.

The $\boxminus$ can be (brutally. Make sure you say in the exam that this solution is not optimal but possible in the name of simplicity) defined as the difference of unit vectors.

4 Possible questions

1. Structure of H matrix (color the blocks.),
2. what if we don't know the position of the observer?
3. dimension of the Jacobian.